Assignment-1

Question 4

Create the following **users, groups, and group memberships** on the system:

- Create a **group** named **sysadmin**.
- Create a **user natasha** who belongs to **sysadmin as a secondary group**.
- Create a **user sarah** who also belongs to **sysadmin as a secondary group.**
- Create a **user harry** who:
  - **does not have access to an interactive shell** on the system.
  - **is not a member of the sysadmin group.**

All three users **natasha, sarah, and harry** should have the password: **Atenorth**

Answer:

```
[root@localhost ~]# groupadd sysadmin
[root@localhost ~]# useradd natasha
[root@localhost ~]# useradd sarah
[root@localhost ~]# useradd harry
[root@localhost ~]# usermod -aG sysadmin natasha
[root@localhost ~]# usermod -aG sysadmin sarah
[root@localhost ~]# usermod -s /sbin/nologin harry
[root@localhost ~]# passwd natasha
Changing password for user natasha.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost ~]# passwd sarah
Changing password for user sarah.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost ~]# passwd harry
Changing password for user harry.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost ~]# passwd natasha
Changing password for user natasha.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
```

Then to verify: cat /etc/passwd → getent group sysadmin

Assignment-2

Question 7

Create a **collaborative directory** /common/admin with the following characteristics:

- The **group ownership** of /common/admin must be **sysadmin.**
- The directory must be **readable, writable, and accessible only to members of the sysadmin group.**
- Other users should **not have access** to this directory.
  (It is understood that **root** has access to all files and directories on the system.)

Answer:

```
[root@localhost ~]# mkdir -p /common/admin
[root@localhost ~]# ls -ld /common/admin
drwxr-xr-x. 2 root root 6 Sep 27 18:15 /common/admin
[root@localhost ~]# chown :sysadmin /common/admin
[root@localhost ~]# chmod 070 /common/admin
[root@localhost ~]# ls -ld /common/admin
d---rwx---. 2 root sysadmin 6 Sep 27 18:15 /common/admin
[root@localhost ~]#
```

Assignment-3

Search for the string "**strato**" in the file:

/usr/share/dict/words

Save the search result in the file: /searchfile

Answer:

```
[root@localhost ~]# grep -iw "strato" /usr/share/dict/words > /searchfile
[root@localhost ~]# cat /searchfile
strato-
strato-cirrus
Strato-cumulus
strato-cumulus
[root@localhost ~]#
```

Here, -w for exact word match

Assignment-4

Create a **user account** with the following requirements:

- Create a user named **barry**.
- The **User ID (UID)** of this user must be **2112**.
- Set the password for the user **barry** to:

  Atenorth

After creating the user, verify that the **UID is correctly assigned.**

Answer:

```
[root@localhost skel]# useradd barry
[root@localhost skel]# usermod -u 2112 barry
[root@localhost skel]# id barry
uid=2112(barry) gid=1011(barry) groups=1011(barry)
[root@localhost skel]# passwd barry
Changing password for user barry.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost skel]# id barry
uid=2112(barry) gid=1011(barry) groups=1011(barry)
[root@localhost skel]#
```

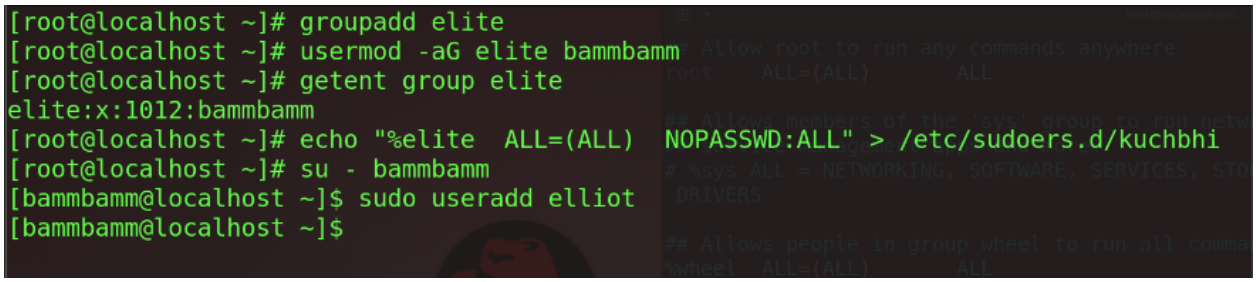
Assignment-5

Configure the **sudo privileges** with the following requirements:

- Ensure a group named **elite** exists on the system.
- Members of the **elite** group must be able to execute **administrative commands using sudo.**

After configuration, verify that members of the **elite** group can execute sudo commands without entering a password.

Answer:



Assignment-6

Configure the system with the following **password policy requirement**:

- Set the **maximum password expiry** to **20 days** for **all newly created users.**
- Ensure that this policy is **automatically applied to every new user account** created on the system.

After completing the configuration, create a test user and verify that the **maximum password expiry is set to 20 days** for that user.

Answer:

1. vim /etc/login.defs
2. Change value of **PASS_MAX_DAYS** to 20

```
[root@localhost ~]# vim /etc/login.defs
[root@localhost ~]# useradd testuser
[root@localhost ~]# chage -l testuser
Last password change                                    : Sep 27, 2026
Password expires                                        : Oct 17, 2026
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 20
Number of days of warning before password expires       : 7
[root@localhost ~]#
```

3. Then use chage command to verify